# CadastrAR: Collaborative Mixed-Reality for Cadastral Field Decision Support

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## Introduction

Cadastral data in New Zealand defines ownership, boundary positions, subdivision rights, and encumbrances across the country's land-information stack, maintained authoritatively through LINZ's Landonline system [1]. Cadastral workflows rely heavily on interpreting boundaries through 2D plans and desktop-based GIS environments. Yet access to that data is not the same as usable knowledge in the field, where decision-critical information remains effectively invisible to the practitioners who must act on it. The reliability of a recorded boundary position is embedded in dataset metadata with no spatial expression on the ground. Known information about subsurface conditions, where it exists in accessible datasets, is similarly absent from the practitioner's view at the moment excavation or construction decisions must be made. Parcel boundaries cannot be confirmed as spatially consistent against the physical environment without manual interpretation of plans that may themselves carry unquantified positional uncertainty [2]. The core limitation isn't data availability or quality, but the absence of systems that make authoritative data visible, legible, and actionable at the moment of decision.

Boundary decisions span multiple practitioners including field operators, surveyors, and planners who reason about the same geometry through disconnected representations yet adjustments propagate consequences across parcels and constraints without being surfaced in a shared spatial context at the moment of decision.

Prior research has made substantial progress toward addressing this problem. Three-dimensional cadastral data representation has advanced practitioners' ability to reason about complex boundary relationships, particularly for strata titles and overlapping rights [3, 4, 5]. Work on augmented reality has brought cadastral geometry closer to the physical site, showing that in-situ boundary visualisation helps practitioners identify discrepancies that plans alone do not reveal [6, 7, 8, 17]. Recent conceptual work has outlined how collaborative AR and VR might restructure survey workflows more broadly [9]. Collaborative MR research in other domains has established that shared spatial presence, multiple participants anchored to the same scene, measurably reduces the referential ambiguity that drives coordination failure in spatially complex tasks [10, 11]. This body of work demonstrates that richer spatial representation of cadastral data is both technically feasible and practically motivated, yet gaps remain in making authoritative data field-legible at the point of decision, and in enabling shared spatial understanding among practitioners before decisions are committed.

CadastrAR addresses these problems as a working mixed-reality prototype by making decision-relevant cadastral information, including boundary geometry, positional reliability, parcel measurements, and constraint status, spatially present and interpretable at the site where decisions are made. It represents the cadastre as a topology-aware network, so that proposed edits propagate immediately and consistently across adjacent parcels, providing a live and coherent view of resulting changes. By anchoring role-differentiated collaboration, including field operator and remote expert, instructor and trainee, and distributed co-reviewers, to a shared spatial scene, it enables practitioners to evaluate boundary proposals and constraint conflicts collectively before they become formal commitments. This transforms cadastral records from abstract coordinates, metadata, and legal descriptions into spatially grounded information that can be directly read, interrogated, and acted upon in situ.

# The CadastrAR System

CadastrAR is a working prototype that represents the cadastre as a network of marks and segments, in which parcel areas, shared boundary edges, adjacency relationships, and segment lengths are structurally coupled. Any proposed edit propagates immediately across all affected parcels, keeping measurements consistent without a separate recalculation step. Because boundary adjustments carry changes across every parcel sharing an edge, practitioners evaluating a proposal see those changes in full rather than reconstructing them manually after the fact [3, 5].

Boundary elements are spatially anchored so that users move through the data at true scale. Marks, segments, and parcel labels are placed in physical space, and positional reliability is communicated through spatial indicators rather than metadata, making data confidence directly interpretable in the scene. Rule-based feedback flags edits that introduce geometric inconsistencies or violate constraints before they compound, supporting correction at the moment of decision rather than at formal submission. The system ingests structured cadastral data consistent with Landonline-format sources [1], grounding the scene in authoritative records rather than reconstructed geometry.

Multi-user collaboration is structured around differentiated roles. A session host maintains the authoritative scene state, while participants join as guests with a synchronised view. This reflects the asymmetry of professional authority over cadastral data. A licensed expert proposing a boundary change and a field operator or trainee receiving guidance occupy different roles, and the interface makes this explicit. Because all participants share the same spatial anchor, they reference the same marks and segments without ambiguity, reducing the referential overhead that makes remote coordination over cadastral geometry costly in current practice [11, 14].

## Development Approach

CadastrAR has been developed through iterative prototype cycles in which each build was exercised with practitioners and experts from cadastral surveying, land information, and geospatial workflows. Their feedback directly steered subsequent builds [9, 16]. Ongoing priorities are oriented toward NZ cadastral practice: authoritative LINZ data ingest, nationally consistent coordinate handling, survey-appropriate snapping and topology validation, subsurface data integration, edit traceability, and outdoor spatial anchoring for field deployment. A two-phase evaluation is planned: the first with novice participants assessing in-situ boundary understanding and constraint reasoning; the second with licensed surveyors assessing fit with professional workflows and requirements for operational deployment.

## Operational Scenarios

Expert engagement during development identified three scenarios:

**Training.** Communicating cadastral geometry to trainees through plans alone requires significant translation between abstract notation and physical space [12, 13]. In CadastrAR, an instructor and trainees share a single anchored scene: the instructor manipulates marks, walks boundary lines, and proposes edits while every participant sees the same topology update in real time. Bearings, lot extents, and easement corridors become spatially legible without requiring interpretation of a 2D diagram, and trainees rehearse boundary reasoning against real spatial relationships without acting on a live dataset.

**Remote Guiding.** When a licensed professional cannot be on site, current practice relies on verbal description of spatial geometry across disconnected representations, prone to misinterpretation. CadastrAR places a remote expert and an on-site operator in a shared anchored view from different physical locations. The expert directs attention to specific marks, proposes adjustments, and negotiates geometry with spatial precision, drawing on the coordination benefits that shared-anchor MR has demonstrated in procedurally complex remote guidance tasks [11, 14].

**Collaborative Verification.** Two or more reviewers jointly traverse parcels, inspect shared edges, and cross-check measurements. Because topology and measurements update live, the resulting changes of a proposed edit surface immediately for all participants, enabling conflicts to be identified and resolved before formal submission rather than discovered after it [5, 15, 16].

# Contributions

1. CadastrAR, a working mixed-reality cadastral prototype in which positional reliability is spatially communicated, boundary edits propagate consistently across a connected parcel network, and measurements remain live throughout interaction.

2. Three synchronised multi-user scenarios for training, remote guidance, and collaborative verification, each targeting a distinct coordination failure in cadastral practice.

3. An iterative, practitioner-informed design process grounded in feedback from cadastral surveying and land-information professionals, and a planned two-phase evaluation targeting in-situ boundary reasoning, constraint interpretation, and collaborative decision-making across novice and professional participants.

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